

Akshaj Kashyap

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EDUCATION

University of California, Santa Barbara

Santa Barbara, CA

B.S. Computer Science | Second-year undergraduate | GPA: 3.76

Expected June 2028

– **Coursework:** Data Structures & Algorithms, Computer Architecture, Probability & Statistics, Systems Programming

EXPERIENCE

Sherwood Lab, UC Santa Barbara

Oct. 2026 – Present

Undergraduate Researcher | Active GPU Memory and Interconnect

Santa Barbara, CA

– **Research focus:** GPU memory hierarchies and interconnects for AI workloads, advised by Professor Tim Sherwood.

PLAXCO Lab

Aug. 2025 – Present

Undergraduate Researcher

Santa Barbara, CA

– Processed 44 electrochemical exports into concentration metrics and plots with Python; continuing to refactor SACMES into reusable analysis code.

Stealth Startup

June 2026 – Sept. 2026

Technology Intern

San Jose, CA

– Built a Python computer vision pipeline with Roboflow, Segment Anything, and depth estimation to detect vegetation near cellular infrastructure for wildfire-risk screening.

Handshake AI Fellowship

Jan. 2026 – Present

AI Evaluation Fellow

Remote

– Completed 20+ rubric-based evaluations of AI-generated image edits, scoring realism and coherence with written justifications.

PROJECTS

CUDA Transformer Runtime | C++20, CUDA, cuBLAS, CMake | [GitHub](#)

- Built a C++20/CUDA runtime for LLaMA-style text generation with BPE tokenization, KV-cached greedy decoding, and custom kernels validated against independent CPU references.
- Passed 21/21 CTest checks; maximum CPU/CUDA logit error was 0.00002563. Generated a 64-token continuation at about 266.8 tokens/s on an RTX 3050 Laptop GPU.

Causal Uplift Experimentation | Python, scikit-learn, FastAPI | [GitHub](#)

- Compared S-, T-, and random-forest learners on synthetic experiments to estimate treatment effects using cross-fitting, bootstrap uncertainty, and policy simulation.
- Packaged a frozen policy and staging API; synthetic trial simulation showed +4.36 percentage points of conversion lift. Kept promotion on hold pending real randomized validation.

Molecular Property Prediction with GNNs | PyTorch Geometric, RDKit, FastAPI | [GitHub](#)

- Benchmarked GCNs, GINs, and Morgan fingerprints for molecular solubility prediction on ESOL scaffold splits; three-seed test RMSE was 1.3395 +/- 0.0738 for GCN vs. 1.8480 +/- 0.0214 for random forest (mean +/- SD).
- Built validation-based model selection and FastAPI/Docker serving with 142 tests. Omitted confidence estimates because ensemble disagreement did not reliably track prediction errors.

LEADERSHIP

SciTrek

Aug. 2025 – Present

Mentor

Santa Barbara, CA

– Helped develop a bioinformatics module used by 100+ K–12 students, translating technical concepts into clear lessons.

C.S.E.P. (F.U.S.E.)

Sep. 2025 – Present

Mentor

Santa Barbara, CA

– Selected as the only undergraduate mentor; helped develop and deliver a STEM module for 250+ students in 9+ classrooms.

TECHNICAL SKILLS

Languages: Python, C++, SQL

ML/Libraries: PyTorch, PyTorch Geometric, scikit-learn, RDKit, NumPy, pandas

ML Systems: CUDA, cuBLAS, CMake, FastAPI

Tools/Platforms: Docker, GitHub Actions, Git, Linux